

MAINTENANCE OPERATIONS CLIENT

Project Overview & Business Problem

Maintenance Operations Analytics & Reporting Automation

Project Attribute	Confirmed Detail
Client	Confidential — presented as Maintenance Operations Client
Engagement Period	Aug–Nov 2024
Role	Business Analyst + Data Analyst
Primary Stakeholder	Maintenance Manager
Data Volume	Approximately 10,000 historical records across 20 teams
Core Problem	Manual consolidation and limited visibility into completion, overdue work, backlog and performance
Solution	Power BI dashboard built from scratch
Confirmed Finding	92% overall completion rate
Confirmed Efficiency Impact	Approximately 3–4 hours of weekly manual reporting effort eliminated
Evidence standard: This document separates confirmed engagement facts from portfolio reconstruction. Unknown historical details are not invented.	

1. Executive Overview

The Maintenance Operations Client engagement addressed a fragmented and time-consuming maintenance reporting process. The Maintenance Manager needed a consolidated view of operational performance, but maintenance updates were distributed across multiple data sources and required manual consolidation.

The engagement combined business analysis and data analysis to structure approximately 10,000 historical records across 20 teams, standardize inconsistent maintenance information, define operational business rules and KPIs, and build a Power BI dashboard from scratch for management reporting.

Confirmed analytical finding: The dashboard surfaced an overall maintenance completion rate of 92%.
Confirmed reporting benefit: The solution eliminated approximately 3–4 hours of weekly manual reporting effort.

The 92% figure is an observed completion-rate finding. It is not presented as a performance improvement caused by the project.

2. Business Context

Maintenance operations generated information across maintenance tasks/work orders, inspections, service requests and equipment-related activity. The operational landscape involved maintenance and field teams, departments, technicians, assets/equipment and locations.

Source information was available through Excel, CSV, Google Sheets and database data. Because these sources were not analysis-ready, management reporting depended on repeated consolidation and cleaning before meaningful operational views could be produced.

3. Business Problem

Problem statement: Maintenance reporting was fragmented and manually consolidated, making it difficult for the Maintenance Manager to obtain timely, consistent visibility into completion, overdue work, backlog and operational performance.

The confirmed problem can be decomposed into four connected areas:

Problem Area	Observed Business Challenge
Reporting effort	Maintenance updates required manual consolidation for recurring reporting.
Performance visibility	Management needed clearer visibility into completion, team performance and technician performance.
Exception visibility	Overdue work, backlog and high-priority pending work were difficult to monitor consistently.
Data consistency	Multiple sources contained duplicates, missing values, inconsistent team names, date formats, status names and blank records.

4. Business Objectives

- Create a consolidated analytical view of maintenance operations.
- Clean and structure approximately 10,000 historical records across 20 teams.
- Standardize operational fields and categories sufficiently for reliable analysis.
- Define consistent maintenance KPI and business-rule logic.
- Provide visibility into completed, open, pending and overdue work.
- Analyze maintenance performance by team and technician.
- Provide priority, asset/equipment and maintenance-category breakdowns.
- Identify backlog, workload imbalance, high-priority pending work, recurring categories and equipment-related issues.
- Replace repetitive manual reporting effort with an interactive Power BI reporting layer.
- Enable management consumption through walkthrough, presentation, documentation and training.

5. Stakeholder & User Need

Stakeholder	Need	How the Project Addressed It
Maintenance Manager	Consolidated operational visibility	Power BI management dashboard
Maintenance Manager	Consistent completion and overdue logic	Defined KPI/business-rule framework
Maintenance Manager	Team and technician performance visibility	Operational performance breakdowns
Maintenance Manager	Visibility into exceptions and backlog	Overdue, pending, priority and backlog analysis
Maintenance Manager	Reduced recurring reporting effort	Structured data and reusable dashboard reporting

Delivery: Confirmed delivery included a live walkthrough, presentation, Power BI dashboard, documentation and training.

6. Data Landscape

Dimension	Confirmed Information
Volume	Approximately 10,000 historical records
Coverage	20 teams
Source types	Excel, CSV, Google Sheets and database
Operational record types	Maintenance task/work order, inspection, service request and equipment-related record
Operational dimensions	Teams, departments, technicians, assets/equipment and locations
Time context	Historical data

7. Available Data Fields

Field	Business Use
Task / Work Order ID	Operational record identification
Team	Team-level performance analysis
Technician	Technician-level activity/performance analysis
Asset / Equipment	Asset/equipment breakdown
Created Date	Work initiation and ageing context
Due Date	Due/overdue monitoring
Completion Date	Completion-time analysis
Status	Completed/open/pending/overdue classification support
Priority	Priority breakdown and high-priority pending visibility
Category	Maintenance-category analysis
Location	Operational location breakdown
Remarks	Supporting operational context

8. Data Quality Problem

The data required cleansing and structuring before KPI calculations and dashboard reporting could be treated as reliable.

Confirmed Data Issue	Analytical Risk
Duplicates	Could overstate work-order/task counts and distort performance measures.
Missing values	Could prevent correct classification, attribution or timing analysis.
Inconsistent team names	Could split one logical team across multiple reporting categories.
Inconsistent date formats	Could invalidate due-date, completion-time and trend calculations.
Inconsistent status names	Could distort completed/open/pending and overdue logic.
Blank records	Could introduce non-actionable or misleading records into aggregates.

9. Analytical Questions

- What proportion of assigned maintenance tasks is completed?
- How many work orders are completed, open or pending?
- Which work orders are overdue?
- What is the average completion time?
- How does completion/performance differ across teams?
- How does technician performance differ?
- How is work distributed by priority?
- Where does backlog exist?
- Are there workload imbalances?
- Are high-priority items still pending?
- Which maintenance categories recur?
- What patterns are visible by asset/equipment?

10. Confirmed Business Rules

Completion Rate: $\text{Completed Maintenance Tasks} \div \text{Total Assigned Maintenance Tasks} \times 100$

Overdue Work: $\text{Due Date} < \text{Current Date AND Status} \neq \text{Completed}$

These definitions will be carried consistently through the KPI Catalogue, Power BI requirements, reconstructed DAX logic, testing documentation and final report. Additional KPI definitions will be documented carefully and will not be represented as exact historical formulas where the original implementation is not retained.

11. Analysis Scope

Analysis Area	Confirmed Coverage
Work-order status	Open, completed and pending work
Completion	Overall and team-wise completion
Exceptions	Overdue work orders and backlog
Timeliness	Average completion time
Priority	Priority breakdown and high-priority pending work
Team performance	Differences across teams
Technician performance	Technician-level performance analysis
Workload	Workload imbalance
Maintenance categories	Recurring categories
Assets/equipment	Equipment-related patterns/issues

12. Confirmed Findings

The engagement produced management visibility into several operational areas. The following findings are retained at a high level without inventing unsupported counts or rankings.

Finding	Evidence Status
Overall completion rate was 92%	Confirmed quantitative finding
Backlog existed	Confirmed qualitative/analytical finding
Overdue work existed	Confirmed qualitative/analytical finding
Performance differed across teams	Confirmed qualitative/analytical finding
Workload imbalance existed	Confirmed qualitative/analytical finding
High-priority work remained pending	Confirmed qualitative/analytical finding
Recurring maintenance categories were identified	Confirmed qualitative/analytical finding
Equipment-related issues/patterns were identified	Confirmed qualitative/analytical finding

Do not infer: Exact backlog size, overdue percentage, team rankings, technician rankings, priority counts, category frequencies and equipment rankings are not retained and should not be invented.

13. Recommendations Boundary

Recommendations were made to the Maintenance Manager and were implemented. However, the exact historical recommendation wording is not retained. This project will therefore distinguish between confirmed historical facts and any recommendations proposed later for the portfolio reconstruction.

Portfolio rule: Any newly designed recommendation must be labeled as a portfolio-proposed recommendation based on confirmed findings, not as an exact historical client recommendation.

14. Solution Overview

A Power BI dashboard was built from scratch to replace fragmented reporting with a consolidated operational view. The exact historical dashboard page names and layout are not retained.

Solution Layer	Purpose
Data consolidation	Bring maintenance information into a structured analytical form
Data cleansing	Resolve or control quality issues before analysis
Business rules	Create consistent completion and overdue logic
Operational analysis	Expose status, backlog, priority, team, technician, category and asset patterns
Power BI reporting	Provide interactive management visibility
Delivery/adoption	Support usage through walkthrough, presentation, documentation and training

Portfolio dashboard reconstruction: For portfolio purposes, logical dashboard views may be designed for Executive Maintenance Overview; Work Order & Completion Analysis; Team & Technician Performance; and Backlog, Priority & Asset Analysis. These are not claimed as exact historical page names.

15. Tools & Their Role

Tool	Confirmed Project Use / Positioning
Excel	Source handling and analytical support
CSV	Source data format
Google Sheets	Source/working data environment
SQL	Data querying/analysis tool used in the engagement
Python	Data preparation/analysis tool used in the engagement
Power Query	Data transformation tool used in the engagement
DAX	Power BI calculation language used in the engagement
Power BI	Dashboard and management reporting solution

Technical integrity: Exact historical SQL queries, Python scripts, Power Query steps and DAX expressions are not retained. Reconstructed technical artifacts will be explicitly labeled.

16. Business Impact

Impact Area	Confirmed Outcome
Operational visibility	Dashboard surfaced an overall 92% completion rate and provided consolidated maintenance-performance visibility.
Reporting efficiency	Approximately 3–4 hours of weekly manual reporting effort were eliminated.
Management reporting	Maintenance information became available through a reusable Power BI reporting layer.
User enablement	Solution was supported by live walkthrough, presentation, documentation and training.

Attribution boundary: The 92% completion rate is a finding, not an improvement caused by the project. The 3–4 hour benefit should not be attributed to a specific person because that detail is not retained.

17. End-to-End Project Flow

Workflow: Fragmented maintenance reporting → requirements/data discovery → consolidation of ~10,000 historical records across 20 teams → cleansing and standardization → KPI/business-rule definition → operational analysis → Power BI dashboard → management walkthrough, documentation and training → ~3–4 hours of weekly manual reporting effort eliminated.

This flow is the central narrative for the portfolio. It demonstrates that the work was not merely dashboard creation; it involved business problem definition, data-quality management, operational KPI design, analytical interpretation, reporting automation and stakeholder enablement.

18. Scope & Evidence Guardrails

Classification	What Can Be Stated
Confirmed	Aug–Nov 2024; BA + DA role; Maintenance Manager; ~10,000 historical records; 20 teams; confirmed sources/fields/data issues; confirmed KPI areas; 92% completion; overdue rule; Power BI built from scratch; delivery activities; recommendations made/implemented at a high level; ~3–4 hours/week reporting effort eliminated.
Portfolio reconstruction	Format document structure, exact logical dashboard page design, proposed data model, reconstructed DAX/SQL, testing specification and any newly formulated recommendations.
Not retained / not claimed	Exact historical dashboard layout, exact source schema, exact code, exact backlog/overdue values, team/technician rankings, exact recommendation wording, formal UAT sign-off, defect counts, production architecture or unsupported operational improvements.

19. Interview-Ready Explanation

90-second narrative: I worked as both a Business Analyst and Data Analyst for a maintenance operations client. The Maintenance Manager was manually consolidating updates from multiple sources and did not have a consistent view of completion, overdue work, backlog or team performance. I worked with approximately 10,000 historical records across 20 teams from Excel, CSV, Google Sheets and database sources. The data had duplicates, missing values, inconsistent team and status names, date-format issues and blank records, so I cleansed and structured it before defining operational reporting logic. I analyzed completed, open, pending and overdue work, completion time, priority, team and technician performance, backlog, workload imbalance, recurring categories and equipment-related issues. I then built a Power BI dashboard from scratch. It surfaced a 92% overall completion rate and gave management consolidated visibility across the operation. I delivered the solution through a live walkthrough, presentation, documentation and training, and the reporting solution eliminated approximately 3 to 4 hours of weekly manual reporting effort.

20. Project Positioning

Case-study positioning: Operational analytics + maintenance reporting automation: converting fragmented historical maintenance data into standardized business rules, management visibility and a reusable Power BI reporting process.